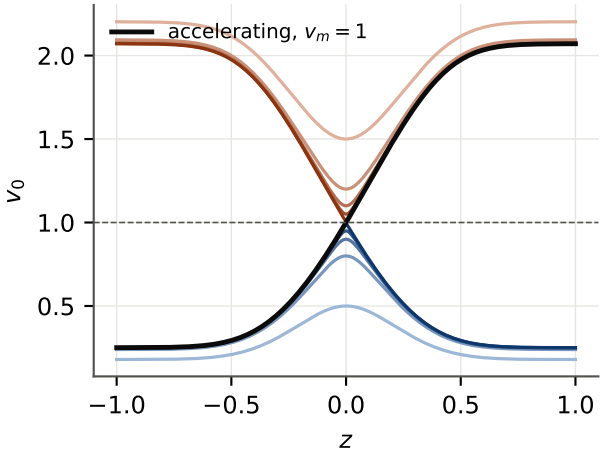
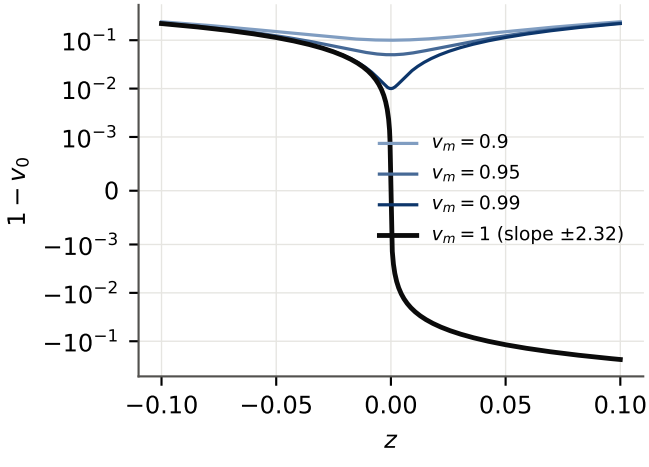

profiles: $v_m = 0.5 \to 0.99$ (blue), $1.5 \to 1.01$ (orange)
accelerating, $v_m = 1$
v_0
2.0
1.5
1.0
0.5
−1.0
−0.5
0.0
0.5
1.0
z


the subsonic family develops a cusp, not the smooth transonic profile
$1 - v_0$
10^{-1}
10^{-2}
10^{-3}
0
-10^{-3}
-10^{-2}
-10^{-1}
$v_m = 0.9$
$v_m = 0.95$
$v_m = 0.99$
$v_m = 1$ (slope ±2.32)
−0.10
−0.05
0.00
0.05
0.10
z